

# PRL GitHub Template

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(Dated: September 11, 2026)

[write ~200 words]

- Catastrophic forgetting in artificial networks has no clean analogue in biological learning, which motivates testing brain-inspired learning rules against it.
- We compare backpropagation, replay, and predictive coding (prospective configuration) on Split-MNIST under matched architecture, data, and stopping criteria.
- The comparison is controlled so that the learning rule is the only free variable, and every reported number is re-derived under the current protocol rather than carried over from earlier exploratory work.

## I. INTRODUCTION

[write ~1000 words]

## II. METHODS

### A. Incremental Learning

Are we saying incremental of continual learning? Be clear about the difference (if any) between the two. Cite literature: van de Ven 2021 looks good. We may also include work from the Vincenzo Lomonaco course material, or other relevant references Why do we care? What function does IL have? How does it relate to machine vs human learning? What are the two scenarios (domain vs class IL), referencing the diagram.

### B. Catastrophic Forgetting

What does catastrophic forgetting look like? Describe figure 901. Talk about the learning/forgetting tradeoff. How does this differ from learning/forgetting in humans? Include a figure or some citations.

### C. The learning rules: Back propagation vs Prospective Configuration

Introduce core learning and "remembering" mechanisms in humans (memory, replay, meta-plasticity, local learning) Cite studies showing that replay etc are well studied. Explain purpose of investigating local learning: Does it contribute to "remembering"? Introduce Back-prop as learning from global error signal, with reference to diagram Introduce prospective configuration as learning from local error signals, with reference to diagram

TABLE I: Every control parameter, and what fixed it. [write ~60 words]

| Parameter     | Value                           | Fixed by     |
|---------------|---------------------------------|--------------|
| Input         | MNIST $14 \times 14$ , $[0, 1]$ | [gap: cite]  |
| Split         | $2 \times 5$ , both scenarios   | [gap: cite]  |
| Hidden width  | $H = 32$                        | 810          |
| Hidden layers | 1                               | 342          |
| Activation    | tanh                            | 802          |
| Output units  | 10 Class-IL, 5 Domain-IL        | structural   |
| Loss          | squared error                   | 812          |
| Optimiser     | SGD, batch 32                   | [gap: cite]  |
| Learning rate | [gap: 0.02, both rules]         | 340, 314     |
| PC settling   | $dt = 0.4$ ; 0.2 off-point      | 334, 347     |
| Stopping      | 90% on both tasks               | section II D |
| Seeds         | 10–19, paired                   | 906          |

Maybe comment on what we might expect to find based on theory Maybe comment on what the literature says about the comparison, citing SB results

### D. Experiment setup and controls

[write ~600 words]

### E. Measuring forgetting

[write ~450 words]

## III. RESULTS

### A. Two scenarios, two forgetting phenotypes

[write ~450 words]

### B. The size of the learning-rule effect

[write ~650 words]

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incremental\_learning\_diagram.png — not generated yet

FIG. 1: Shows the difference between Domain IL and Class IL w.r.t. our setup.

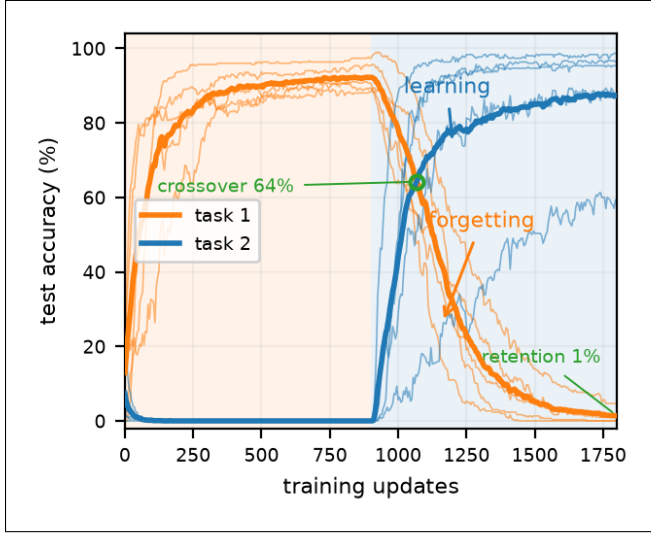


FIG. 2: [write ~110 words]

C. Testing the mechanism prospective configuration is credited with

[write ~500 words]

D. Why these are two investigations, not one

[write ~700 words]

E. Evidence for an output-competition component

[write ~700 words]

#### IV. DISCUSSION

[write ~900 words]

A. What this work does not establish

[write ~500 words]

#### Appendix A: Equilibrium Propagation

[write ~400 words]

- Equilibrium propagation was included in early exploratory work as a second local-learning-rule test case, but is shelved for the current report series to keep scope on the predictive-coding-vs-backprop comparison.
- Preliminary, pre-series work found it never outperformed backprop on retention and showed measurably less efficient credit assignment.
- Nothing in this appendix is offered as evidence under the current protocol; it is retained for context on why the rule was set aside.

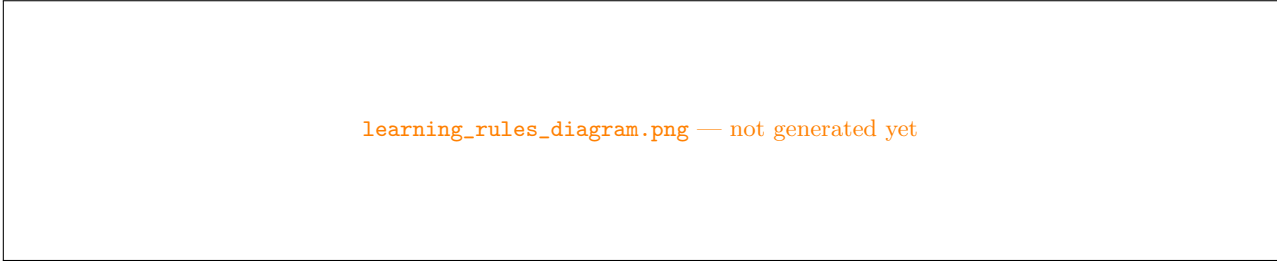
A rectangular box with a thin black border. Inside the box, the text "learning\_rules\_diagram.png — not generated yet" is centered in a light orange or brown color. The box is set against a white background.

FIG. 3: **[write ~110 words]**

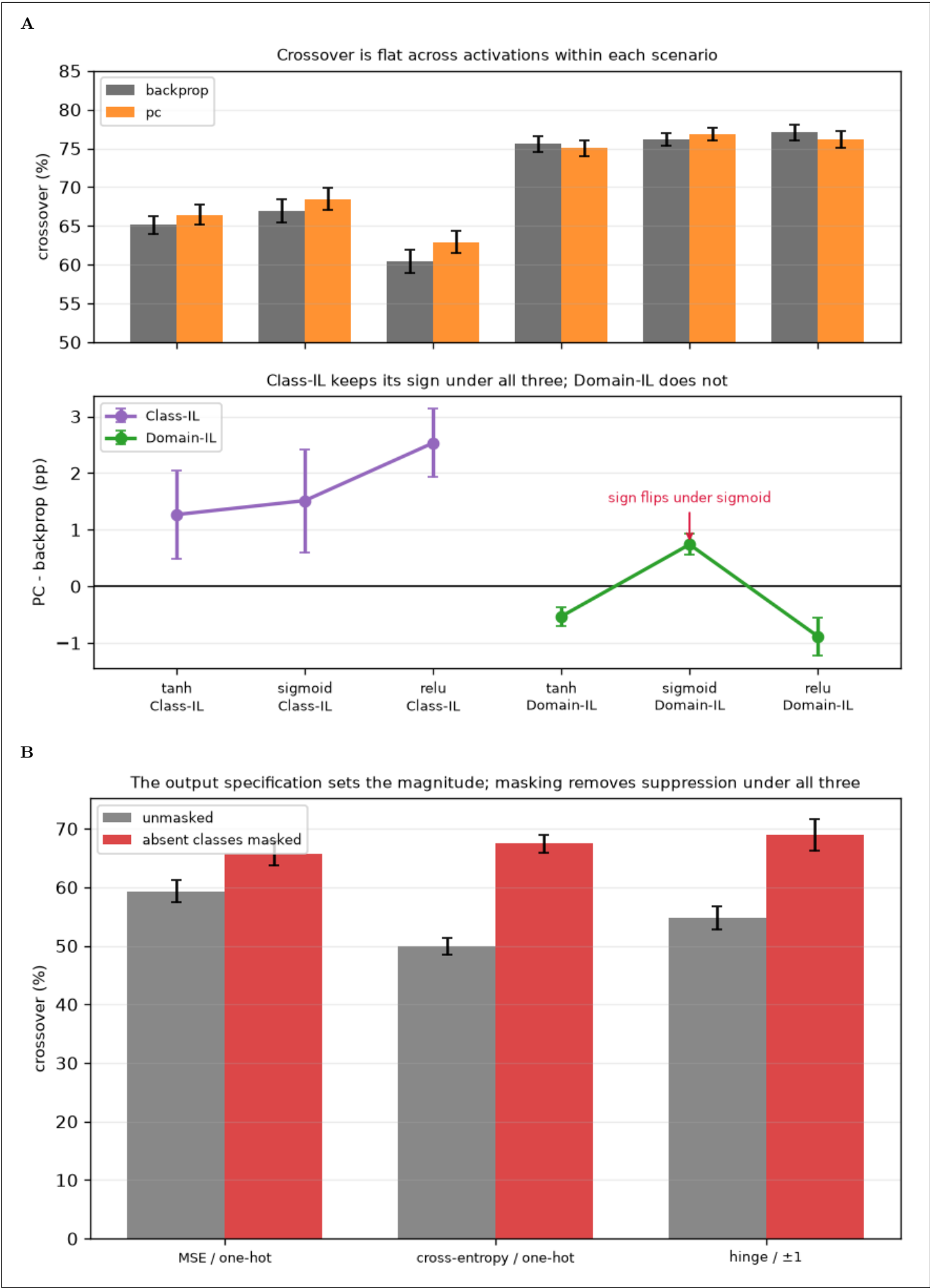


FIG. 4: *[gap: 903 not written; needs 802, which is queued]***[write ~110 words]**

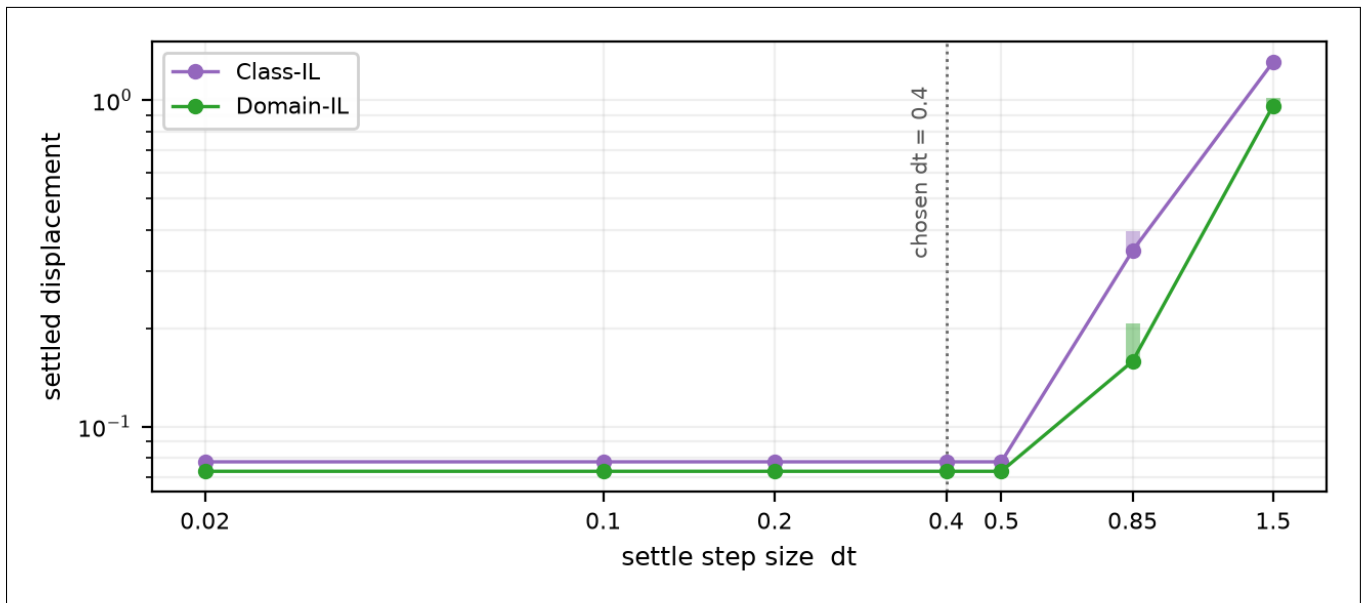


FIG. 5: *[gap: 904 not written; copies script 334]***[write ~110 words]**

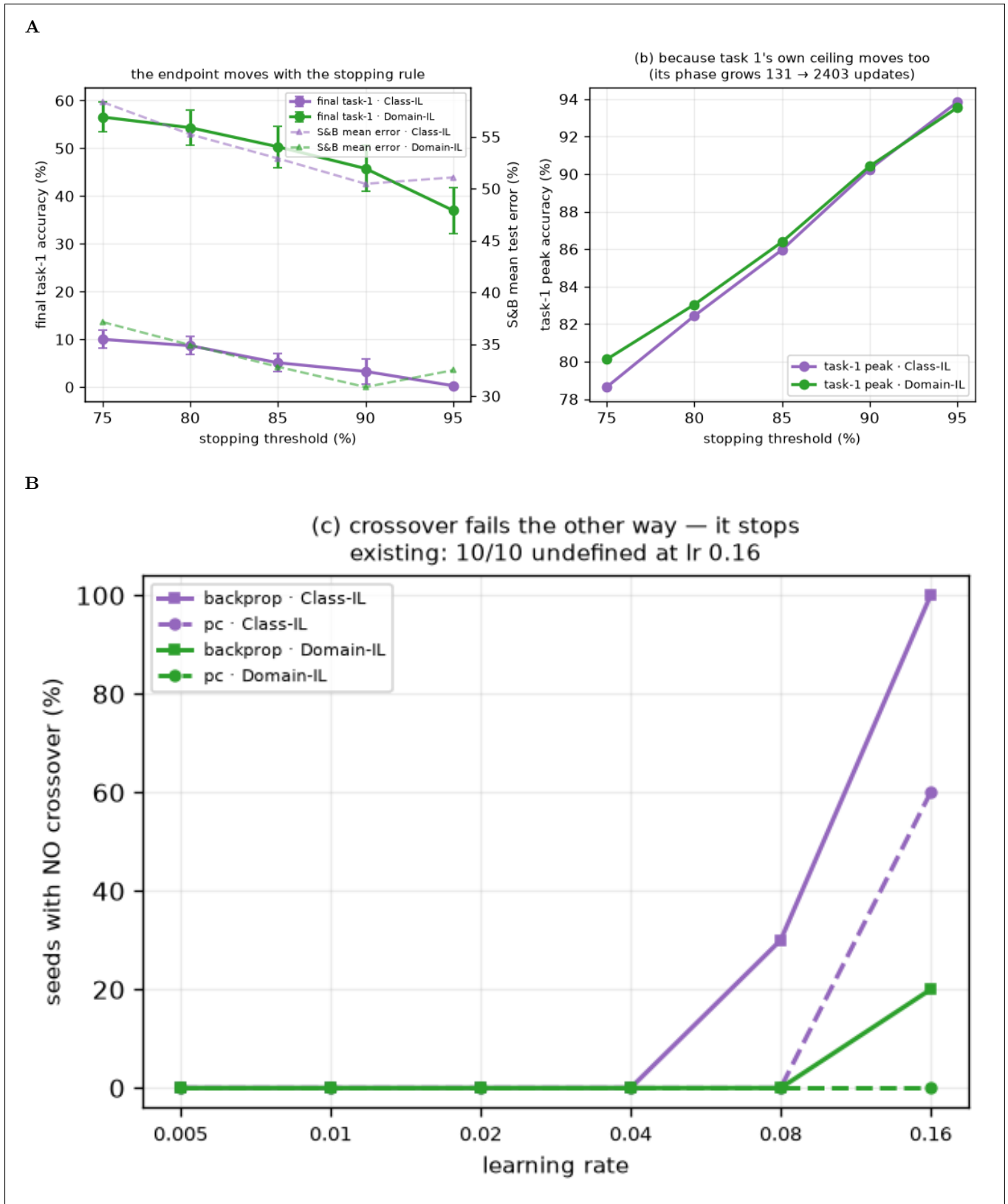


FIG. 6: *[gap: 907 not written; merges 111 and 343]**[write ~110 words]*

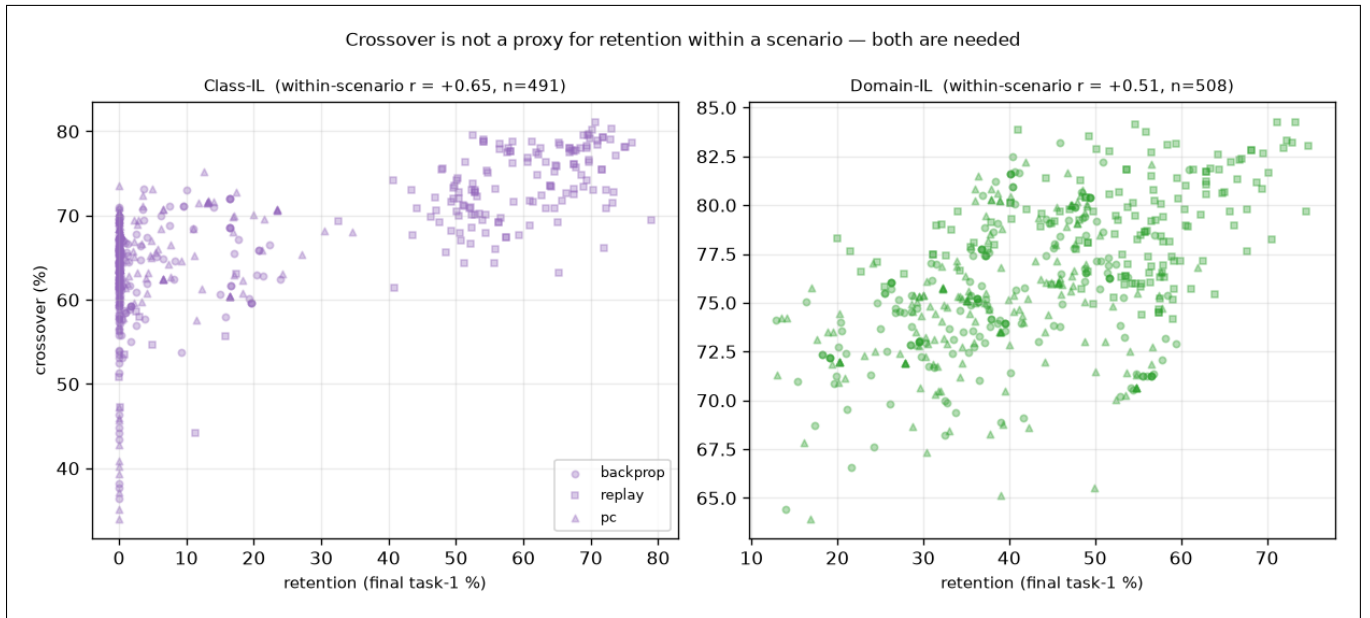


FIG. 7: [write ~110 words]

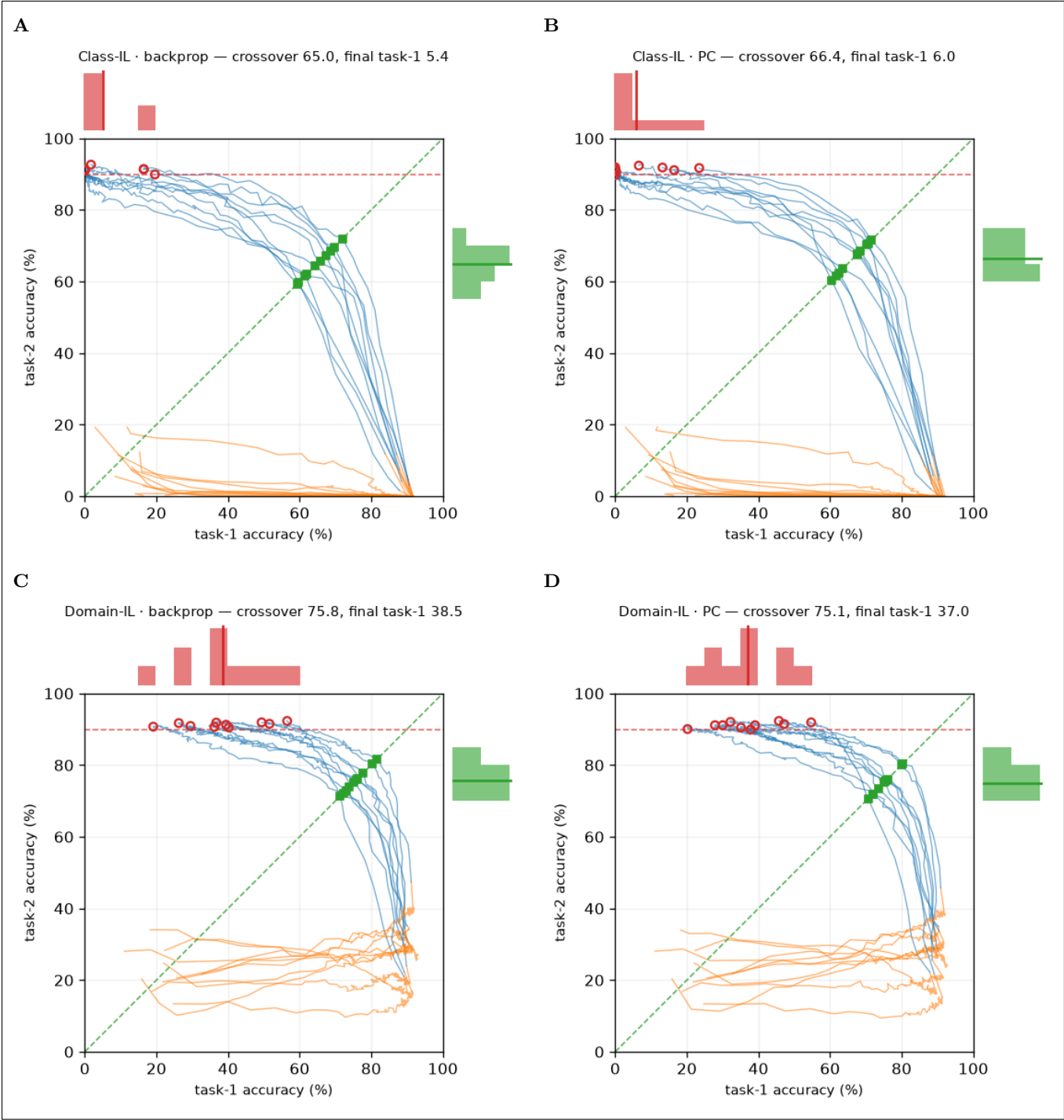


FIG. 8: [write ~130 words]

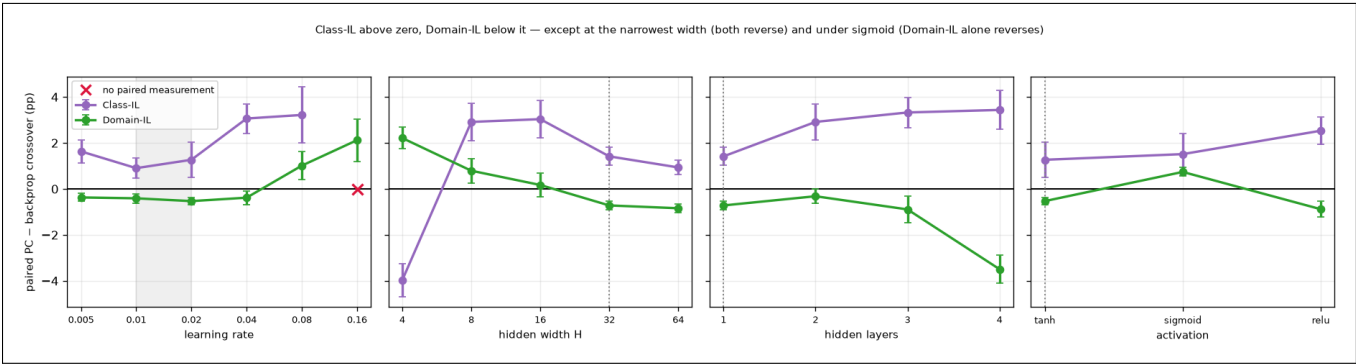


FIG. 9: [write ~150 words]

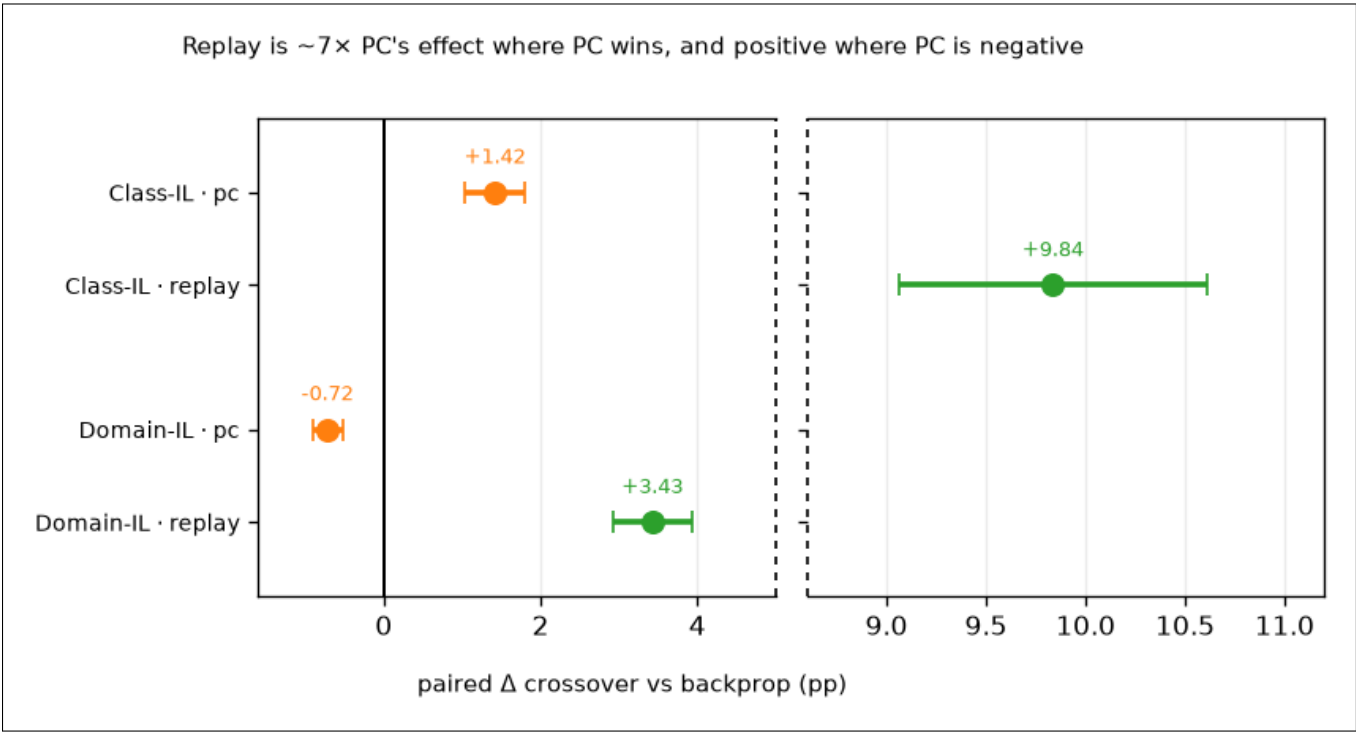


FIG. 10: [write ~50 words]

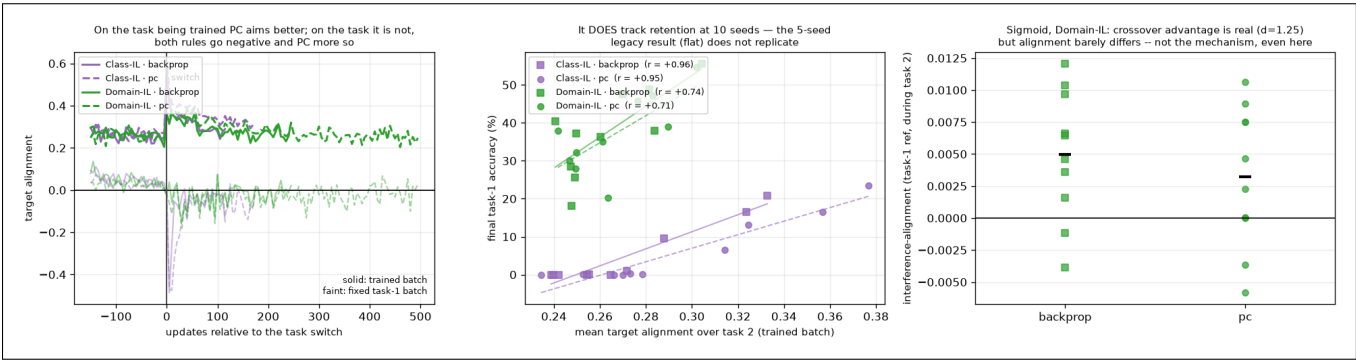


FIG. 11: [gap: 915 not written; 803 has run][write ~110 words]

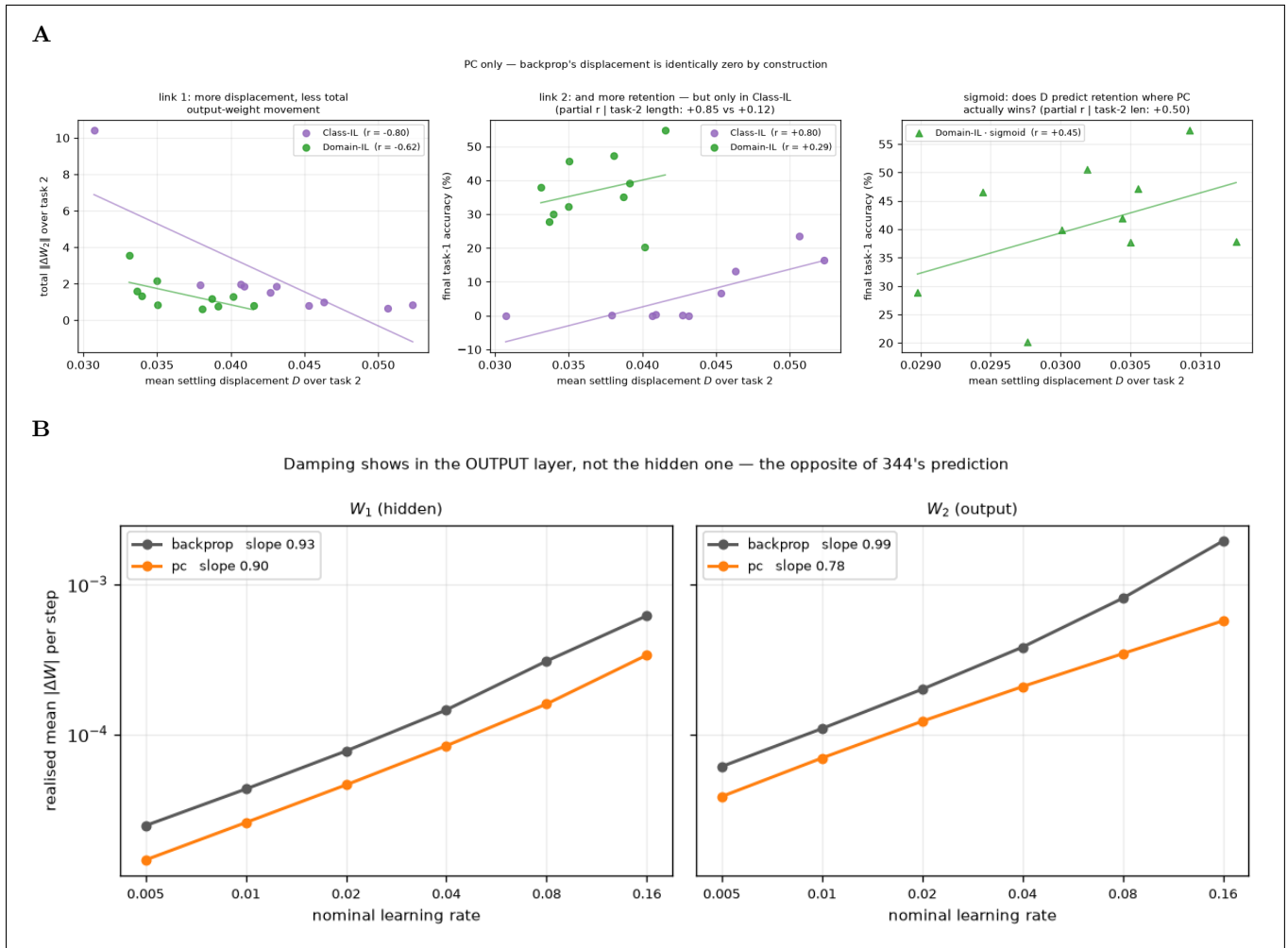


FIG. 12: [gap: 916 not written; 803 has run][write ~110 words]

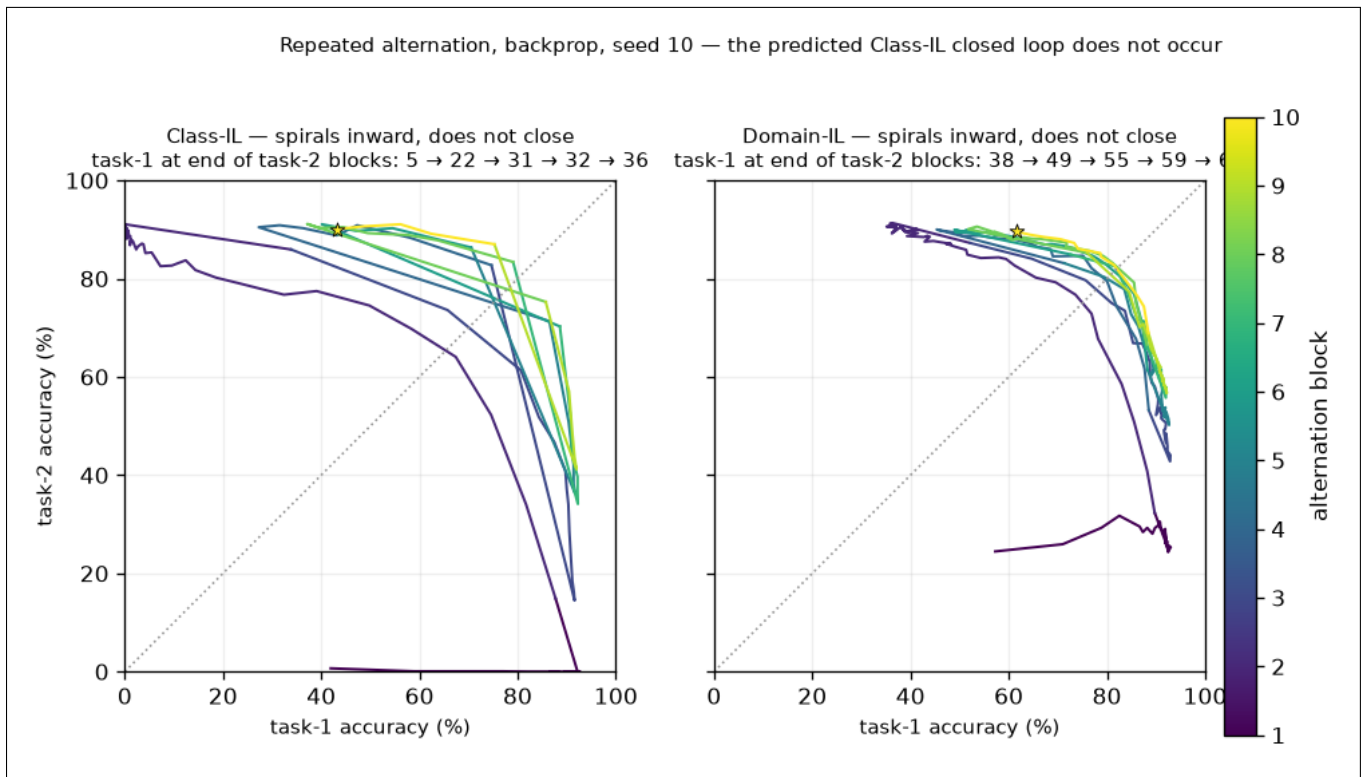


FIG. 13: *[gap: 917 not written; 804 Class-IL done, Domain-IL running]***[write ~150 words]**

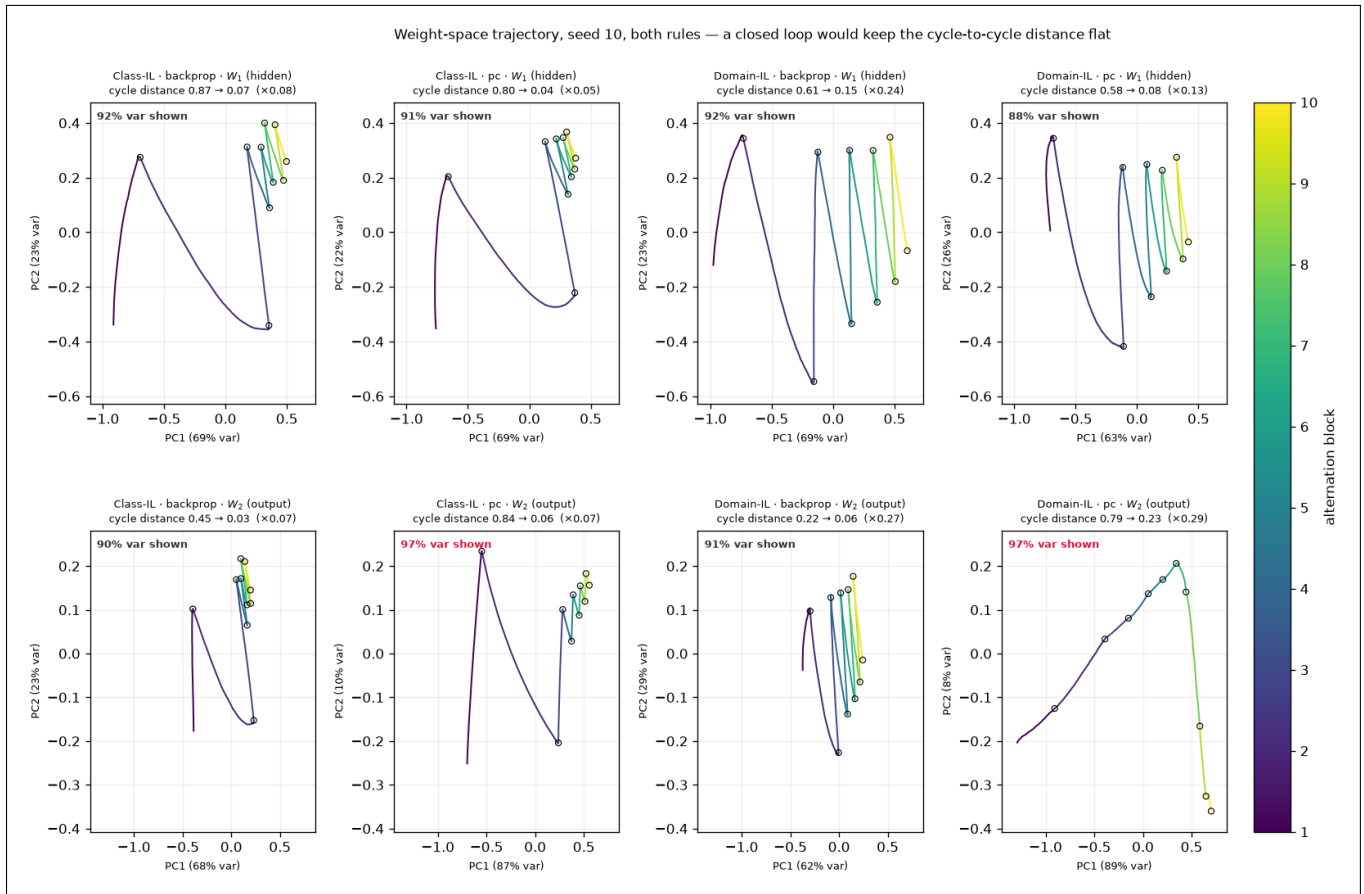


FIG. 14: *[gap: 918 not written; needs 804]***[write ~110 words]**

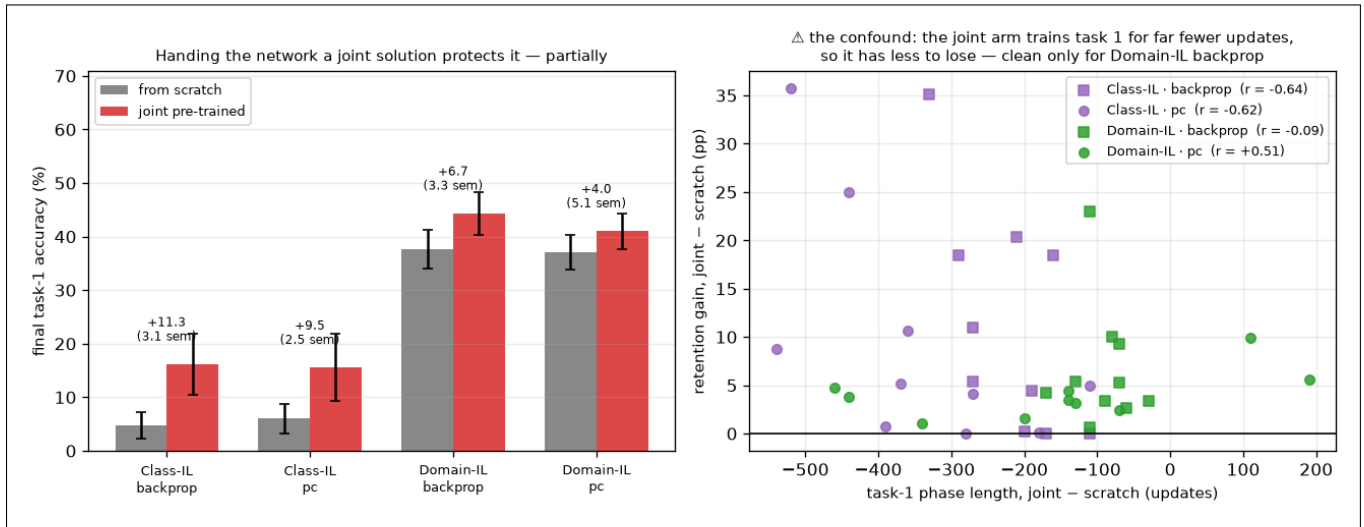


FIG. 15: *[gap: 919 not written; 805 queued]***[write ~110 words]**

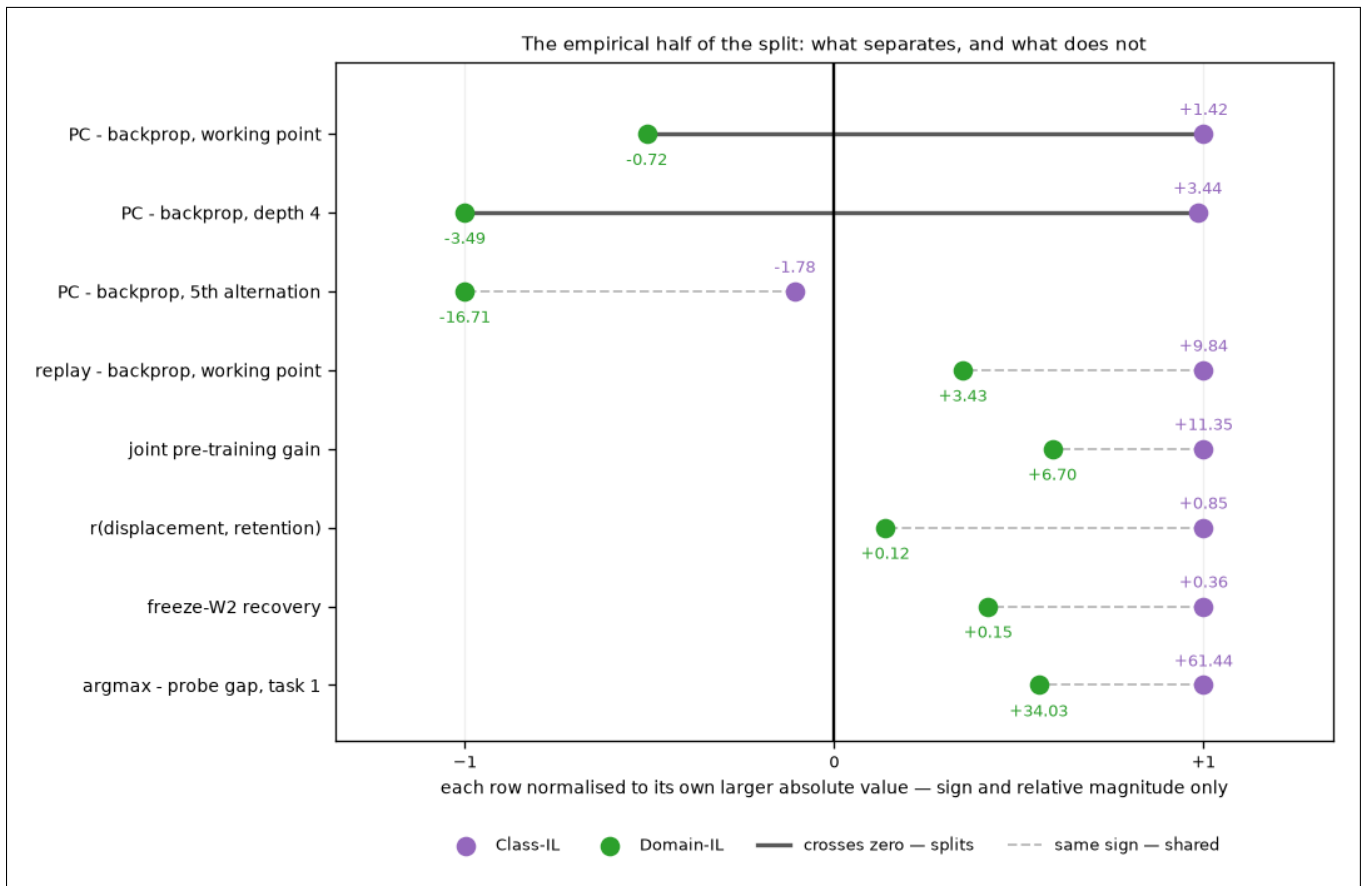


FIG. 16: *[gap: 920 not written; re-analysis only]***[write ~110 words]**

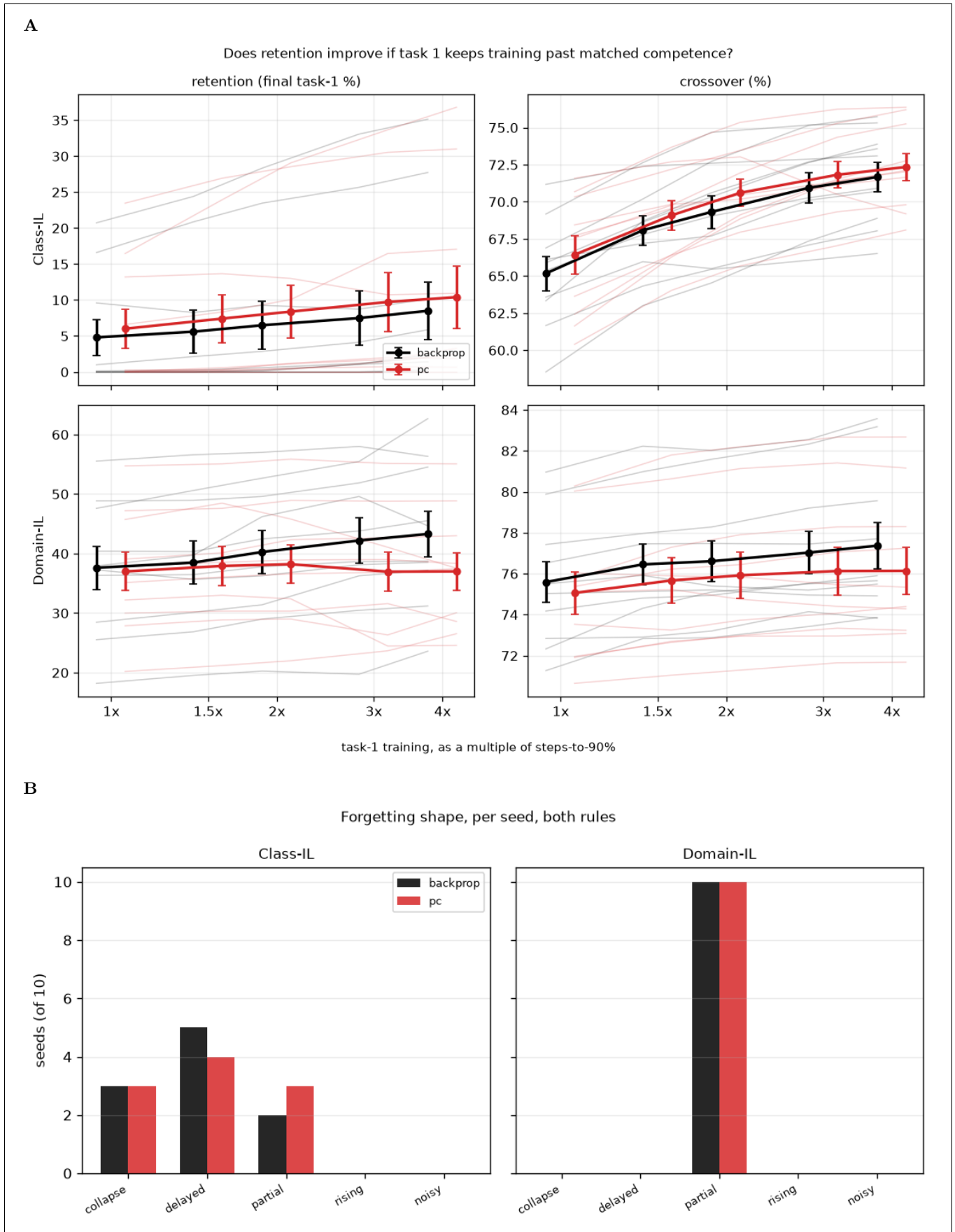


FIG. 17: [write ~130 words]

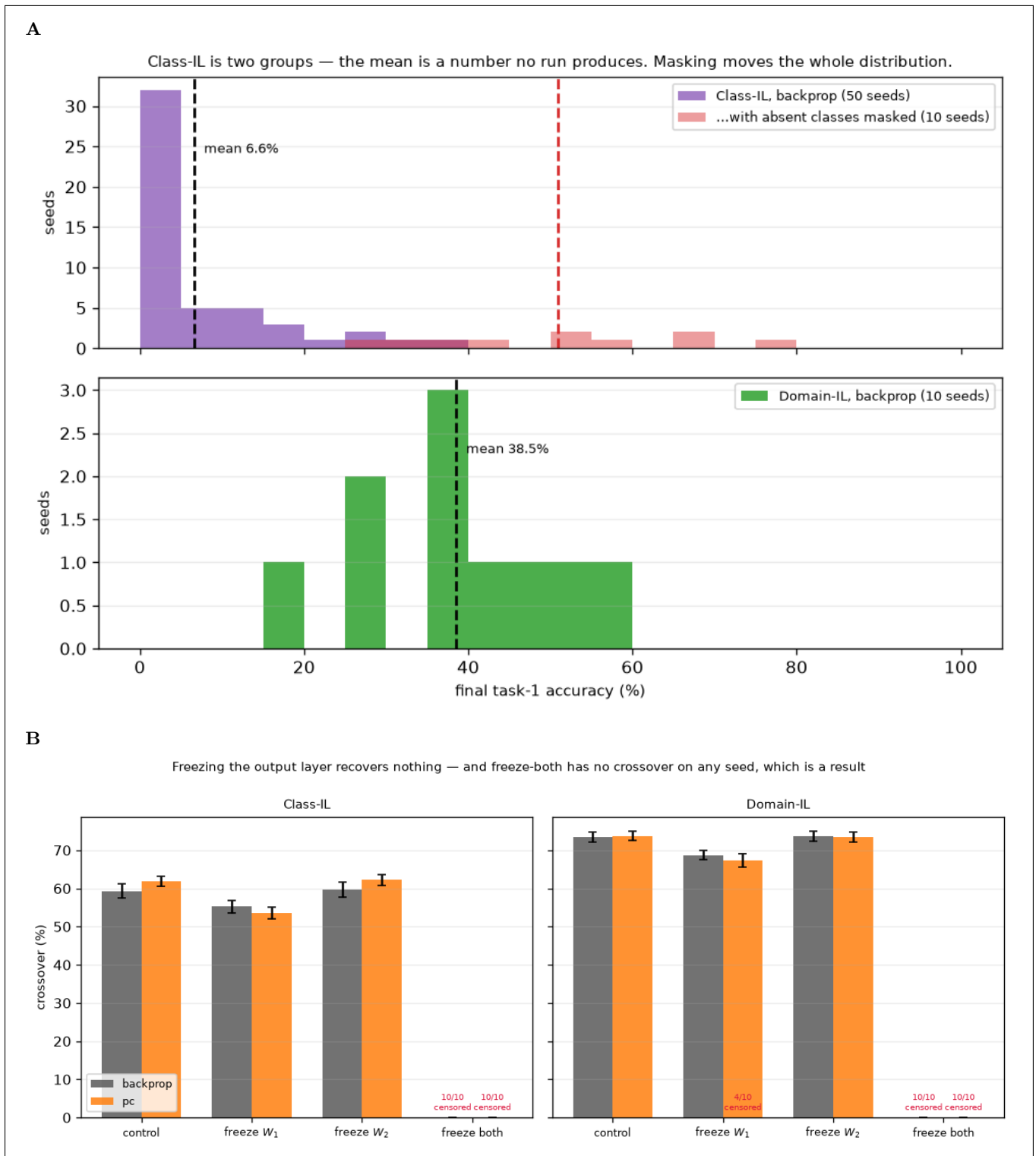


FIG. 18: *[gap: 921 not written]***[write ~110 words]**

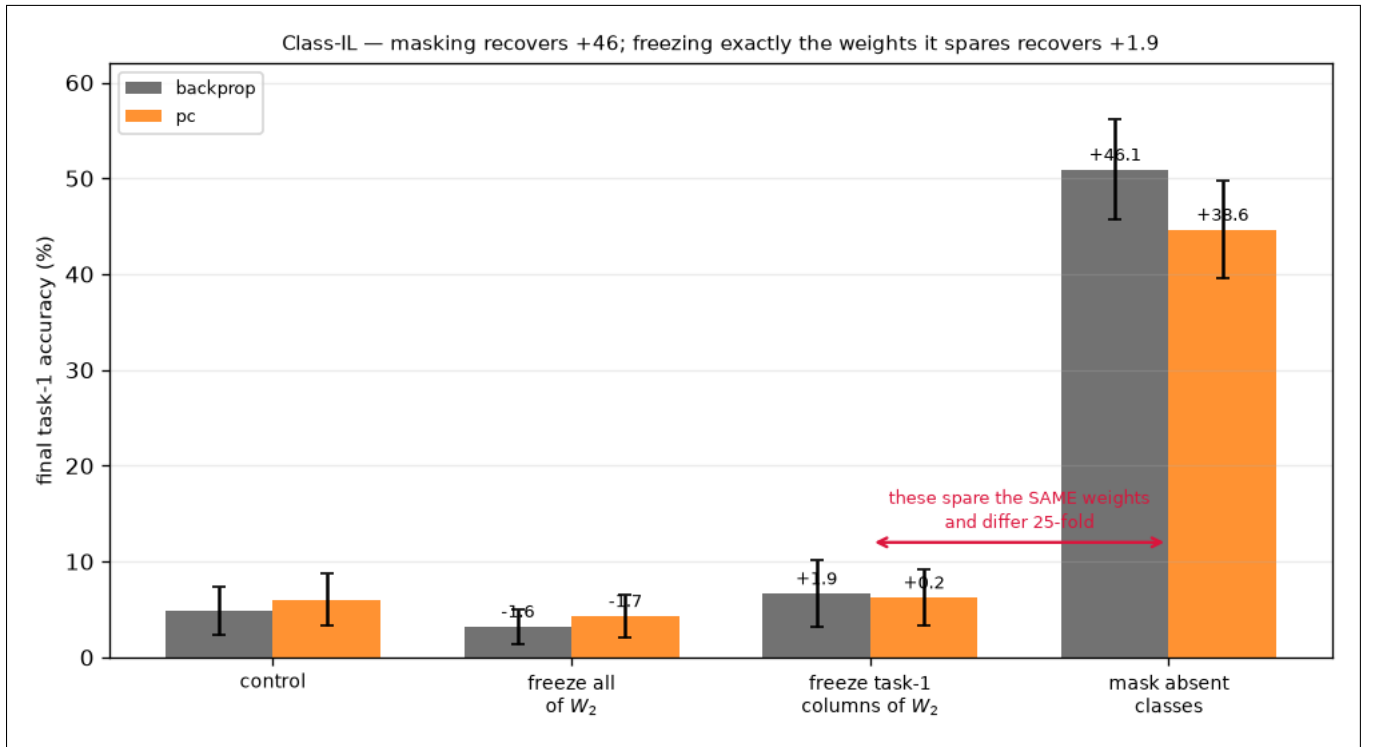


FIG. 19: *[gap: 922 not written; 806 queued]***[write ~110 words]**

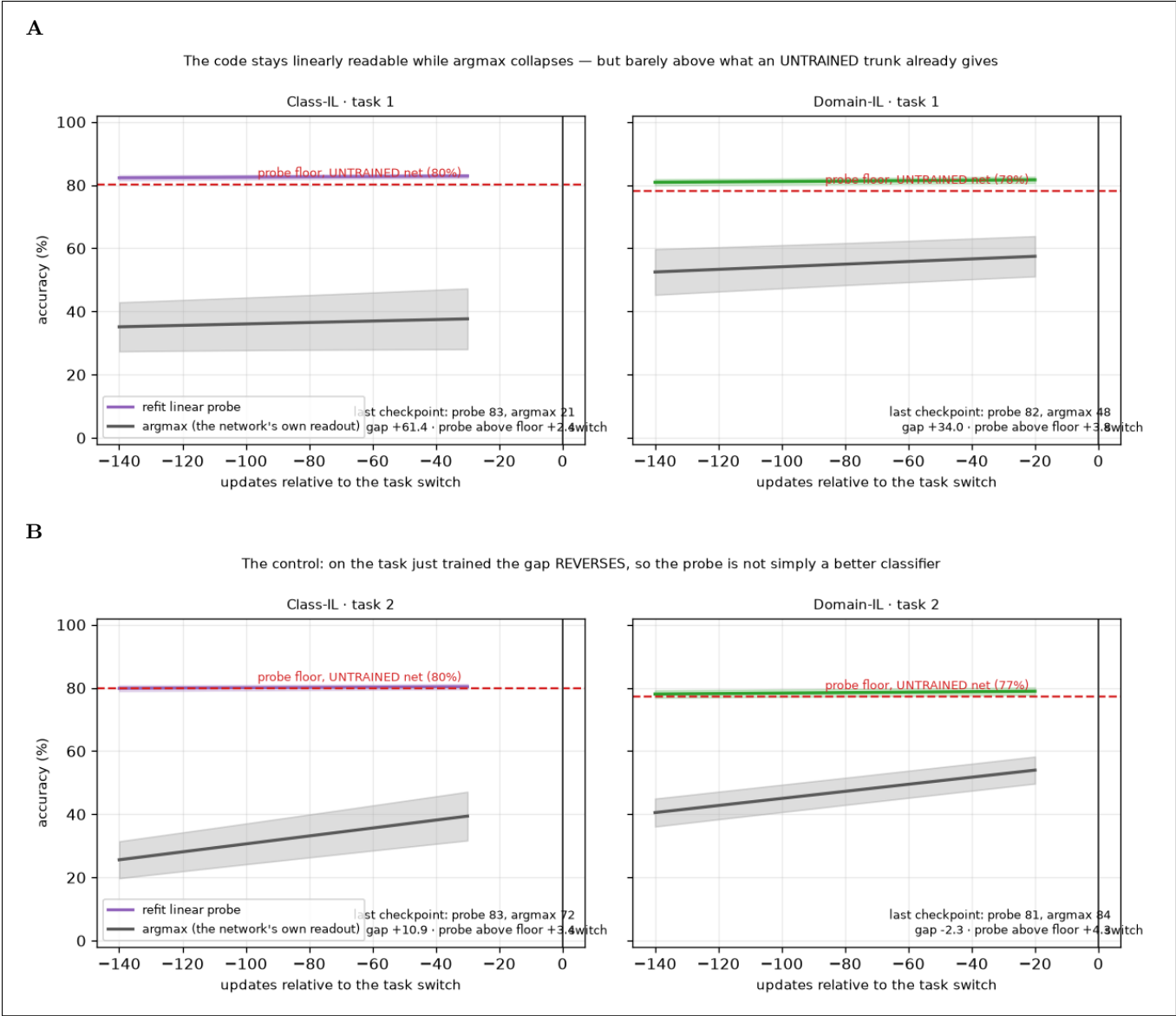


FIG. 20: [gap: 923 not written; 803 has run][write ~150 words]

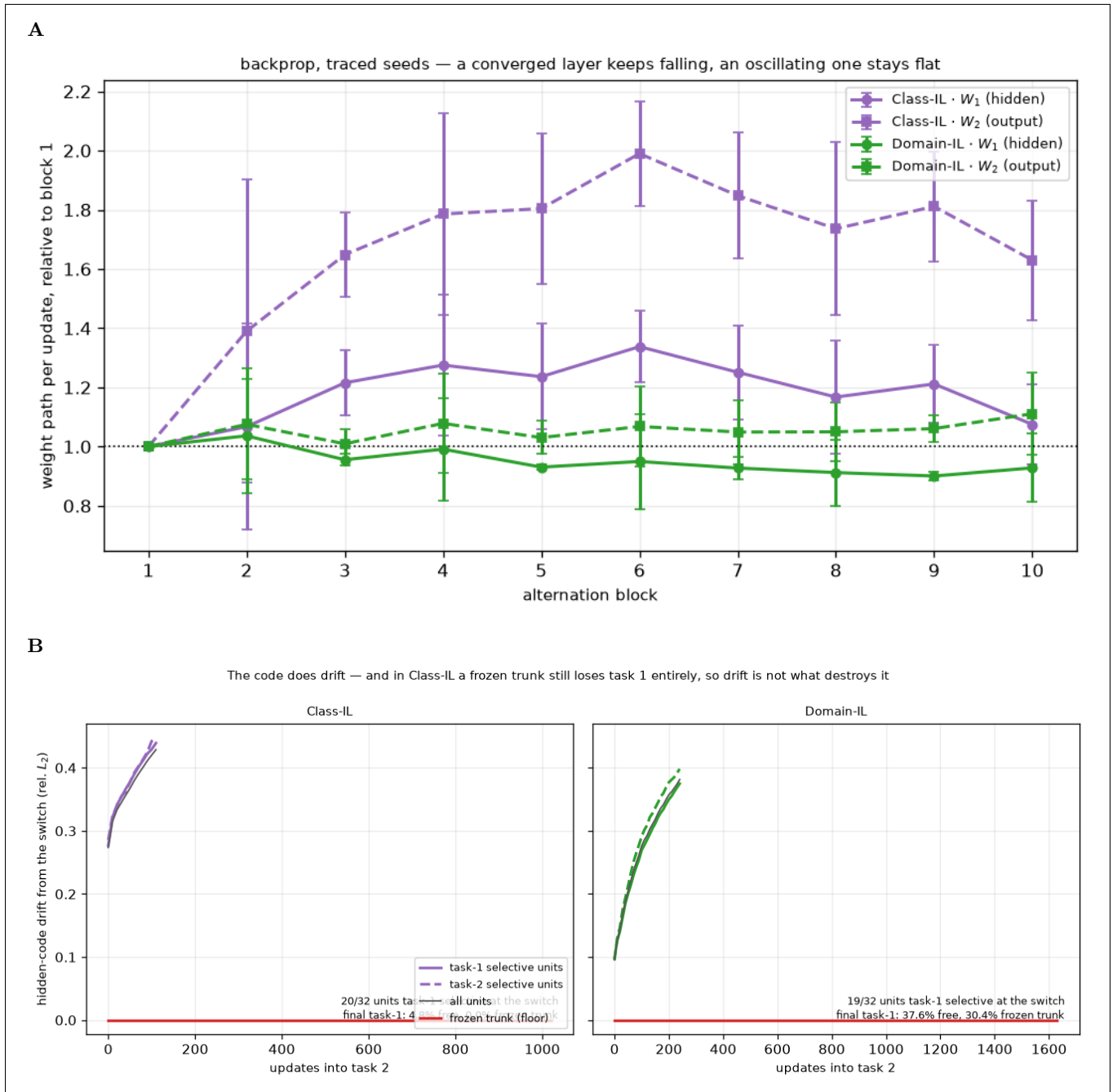


FIG. 21: [gap: 924 not written; needs 803 and 804][write ~110 words]

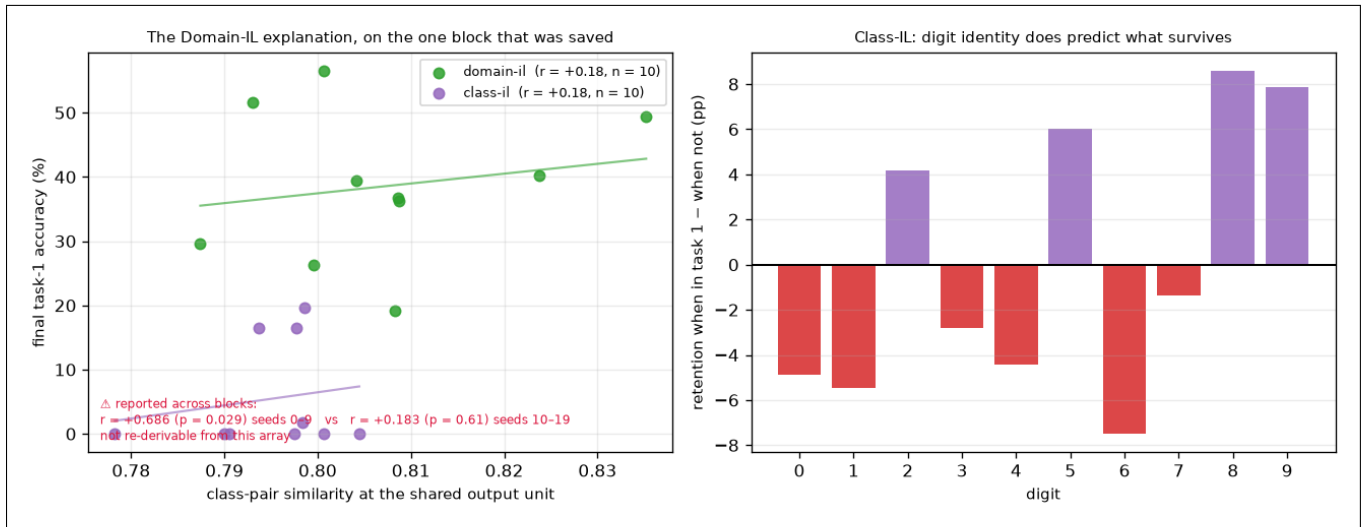


FIG. 22: *[gap: 925 not written]***[write ~110 words]**



FIG. 23: *[gap: 931 not written]***[write ~110 words]**